

Thr Firing Schmitt

Component Design Document

1 Description

Thruster firing Schmitt algorithm converts thruster force commands to on-time commands using Schmitt-trigger (hysteresis) pulse-width modulation.

2 Requirements

No requirements have been specified for this component.

3 Design

3.1 At a Glance

Below is a list of useful parameters and statistics that give a quick look into the makeup of the component.

- **Execution** - *passive*
- **Number of Connectors** - 5
- **Number of Invokee Connectors** - 3
- **Number of Invoker Connectors** - 2
- **Number of Generic Connectors** - *None*
- **Number of Generic Types** - *None*
- **Number of Unconstrained Arrayed Connectors** - *None*
- **Number of Commands** - *None*
- **Number of Parameters** - 5
- **Number of Events** - *None*
- **Number of Faults** - *None*
- **Number of Data Products** - *None*
- **Number of Data Dependencies** - 1
- **Number of Packets** - *None*

3.2 Diagram



Figure 1: Thr Firing Schmitt component diagram.

3.3 Connectors

Below are tables listing the component's connectors.

3.3.1 Invokee Connectors

The following is a list of the component's *invokee* connectors:

Table 1: Thr Firing Schmitt Invokee Connectors

Name	Kind	Type	Return_Type	Count
Tick_T_Recv_Sync	recv_sync	Tick.T	-	1
Reset_Tick_T_Recv_Sync	recv_sync	Tick.T	-	1
Parameter_Update_T_Modify	modify	Parameter_Update.T	-	1

Connector Descriptions:

- **Tick_T_Recv_Sync** - Run the algorithm up to the current time.
- **Reset_Tick_T_Recv_Sync** - Reset the algorithm's Schmitt-trigger hysteresis state. Called on GNC state change.
- **Parameter_Update_T_Modify** - The parameter update connector.

3.3.2 Invoker Connectors

The following is a list of the component's *invoker* connectors:

Table 2: Thr Firing Schmitt Invoker Connectors

Name	Kind	Type	Return_Type	Count
Data_Product_Fetch_T_Request	request	Data_Product_Fetch.T	Data_Product_Return.T	1
Thr_On_Time_Cmd_T_Send	send	Thr_On_Time_Cmd.T	-	1

Connector Descriptions:

- **Data_Product_Fetch_T_Request** - Fetch a data product item from the database.
- **Thr_On_Time_Cmd_T_Send** - Send the computed thruster on-time command directly to the actuation interface component. Only honored by the receiver while the thruster group is in closed-loop control.

3.4 Interrupts

This component contains no interrupts.

3.5 Initialization

Below are details on how the component should be initialized in an assembly.

3.5.1 Component Instantiation

This component contains no instantiation parameters in its discriminant.

3.5.2 Component Base Initialization

This component contains no base class initialization, meaning there is no `init_Base` subprogram for this component.

3.5.3 Component Set ID Bases

This component contains commands, events, packets, faults, or data products that require a base identifier to be set at initialization. The `set_Id_Bases` procedure must be called with the following parameters:

Table 3: Thr Firing Schmitt Set Id Bases Parameters

Name	Type
<code>Parameter_Id_Base</code>	<code>Parameter_Types.Parameter_Id_Base</code>

Parameter Descriptions:

- **Parameter_Id_Base** - The value at which the component's parameter identifiers begin.

3.5.4 Component Map Data Dependencies

This component contains data dependencies that need their IDs mapped to valid data product IDs at initialization. Each data dependency also needs a stale limit provided in microseconds. If the timestamp of the data dependency is older than the reference time (usually the current time) minus the stale limit then the data dependency is considered too old to safely use. The `map_Data_Dependencies` procedure must be called with the following parameters:

Table 4: Thr Firing Schmitt Map Data Dependencies Parameters

Name	Type
<code>Thruster_Force_Cmd_Id</code>	<code>Data_Product_Types.Data_Product_Id</code>
<code>Thruster_Force_Cmd_Stale_Limit</code>	<code>Ada.Real_Time.Time_Span</code>

Parameter Descriptions:

- **Thruster_Force_Cmd_Id** - The external data product ID used to fetch the value for the 'Thruster_Force_Cmd' data dependency.
- **Thruster_Force_Cmd_Stale_Limit** - The stale limit used for the 'Thruster_Force_Cmd' data dependency. If the fetched data product is older than the given `Time_Span` then it is deemed stale and unfit for use by the component.

3.5.5 Component Implementation Initialization

The calling of this implementation class initialization procedure is mandatory. Initializes the thruster firing Schmitt algorithm with the control period and the default parameter values. The `init` sub-program requires the following parameters:

Table 5: Thr Firing Schmitt Implementation Initialization Parameters

Name	Type	Default Value
Control_Period	Basic_Types.Positive_Short_Float	<i>None provided</i>

Parameter Descriptions:

- **Control_Period** - [s] Time between two algorithm updates. Fixed by the rate at which the assembly schedules the component, so it is not a runtime parameter. The type keeps it greater than zero.

3.6 Commands

The Thr Firing Schmitt component has no commands.

3.7 Parameters

Parameters for the Thr Firing Schmitt component

Table 6: Thr Firing Schmitt Parameters

Local ID	Parameter Name	Type	Default Value
0x0000 (0)	Max_Thrust	Packed_F32x8.T	[1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0]
0x0001 (1)	Levels	Levels_On_Off.T	(Level_On=>0.75, Level_Off=>0.25)
0x0002 (2)	Thr_Min_Fire_Time	Packed_F32.T	(Value=>0.02)
0x0003 (3)	On_Time_Saturation_Factor	Packed_F32.T	(Value=>1.0)
0x0004 (4)	Thrust_Pulsing_Regime	Packed_Pulsing_Regime.T	(Value=>Thr_Firing_Remainder_Enums.Pulsing_Regime.On_Pulsing)

Parameter Descriptions:

- **Max_Thrust** - [N] Per-thruster maximum thrust. Every entry must be finite and greater than zero.
- **Levels** - [-] Schmitt-trigger ON and OFF duty cycle fractions
- **Thr_Min_Fire_Time** - [s] Minimum thruster fire time
- **On_Time_Saturation_Factor** - On-time saturation factor
- **Thrust_Pulsing_Regime** - Thrust pulsing regime

3.8 Events

The Thr Firing Schmitt component has no events.

3.9 Data Products

The Thr Firing Schmitt component has no data products.

3.10 Data Dependencies

Data dependencies for the Thr Firing Schmitt component.

Table 7: Thr Firing Schmitt Data Dependencies

Data Dependency Name	Type
Thruster_Force_Cmd	Thr_Force_Cmd.T

Data Dependency Descriptions:

- **Thruster_Force_Cmd** - Thruster force command (8 thrusters)

3.11 Packets

The Thr Firing Schmitt component has no packets.

3.12 Faults

The Thr Firing Schmitt component has no faults.

4 Unit Tests

The following section describes the unit test suites written to test the component.

4.1 *Thr_Firing_Schmitt_Tests* Test Suite

This is a unit test suite for the Thr Firing Schmitt component

Test Descriptions:

- **Test** - Run algorithm to ensure integration is sound.
- **Test_Reset** - Verify the reset connector clears the Schmitt-trigger hysteresis state.
- **Test_Min_Fire_Time_Floor** - A duty cycle at or above Level_On latches the thruster on at the minimum fire time, and one at or below Level_Off latches it off, regardless of the previous state.
- **Test_On_Time_Saturation** - An on-time request that reaches the control period saturates to On_Time_Saturation_Factor times the control period.
- **Test_Off_Pulsing_Offset** - Off-pulsing adds the maximum thrust to the requested force and clamps the sum at zero.
- **Test_Thruster_Independence** - Each thruster carries its own hysteresis state, so the same intermediate force yields different on-times.
- **Test_Pulsing_Regime_Validation** - A byte outside the pulsing regime enumeration is rejected at parameter staging by E8 type validation.

- **Test_Invalid_Parameter** - A staged parameter set the algorithm's config validators would reject is refused by Validate_Parameters, one field at a time.
- **Test_Invalid_Data_Dependency** - Ensure a data dependency with the wrong identifier is treated as a wiring defect and fails the tick's assertion.

5 Appendix

5.1 Preamble

This component contains no preamble code.

5.2 Packed Types

The following section outlines any complex data types used in the component in alphabetical order. This includes packed records and packed arrays that might be used as connector types, command arguments, event parameters, etc..

Data_Product.T:

Generic data product packet for holding arbitrary data types

Table 8: Data_Product Packed Record : 1096 bits (*maximum*)

Name	Type	Range	Size (Bits)	Start Bit	End Bit	Variable Length
Header	Data_Product_Header.T	-	72	0	71	-
Buffer	Data_Product_Types.Data_Product_Buffer_Type	-	1024	72	1095	Header.Buffer_Length

Field Descriptions:

- **Header** - The data product header
- **Buffer** - A buffer that contains the data product type

Data_Product_Fetch.T:

A packed record which holds information for a data product request.

Table 9: Data_Product_Fetch Packed Record : 16 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Id	Data_Product_Types.Data_Product_Id	0 to 65535	16	0	15

Field Descriptions:

- **Id** - The data product identifier

Data_Product_Header.T:

Generic data_product packet for holding arbitrary data_product types

Table 10: Data_Product_Header Packed Record : 72 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Time	Sys_Time.T	-	48	0	47
Id	Data_Product_Types. Data_Product_Id	0 to 65535	16	48	63
Buffer_Length	Data_Product_ Types.Data_Product_ Buffer_Length_Type	0 to 128	8	64	71

Field Descriptions:

- **Time** - The timestamp for the data product item.
- **Id** - The data product identifier
- **Buffer_Length** - The number of bytes used in the data product buffer

Data_Product_Return.T:

This record holds data returned from a data product fetch request.

Table 11: Data_Product_Return Packed Record : 1104 bits (*maximum*)

Name	Type	Range	Size (Bits)	Start Bit	End Bit	Variable Length
The_ Status	Data_ Product_ Enums. Fetch_ Status.E	0 => Success 1 => Not_Available 2 => Id_Out_Of_Range	8	0	7	-
The_ Data_ Product	Data_ Product. T	-	1096	8	1103	-

Field Descriptions:

- **The_Status** - A status relating whether or not the data product fetch was successful or not.
- **The_Data_Product** - The data product item returned.

Levels_On_Off.T:

Schmitt-trigger ON and OFF duty cycle fractions.

Table 12: Levels_On_Off Packed Record : 64 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Level_On	Short_Float	-3.40282e+38 to 3.40282e+38	32	0	31
Level_Off	Short_Float	-3.40282e+38 to 3.40282e+38	32	32	63

Field Descriptions:

- **Level_On** - [-] ON duty cycle fraction
- **Level_Off** - [-] OFF duty cycle fraction

Packed_F32.T:

Single component record for holding packed 32-bit floating point number.

Table 13: Packed_F32 Packed Record : 32 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Value	Short_Float	-3.40282e+38 to 3.40282e+38	32	0	31

Field Descriptions:

- **Value** - The 32-bit floating point number.

Packed_F32x8.T:

Packed 8 element vector of 32-bit floats. Used for thruster command arrays sized to the system thruster count.

Table 14: Packed_F32x8 Packed Array : 256 bits

Type	Range	Element Size (Bits)	Length	Total Size (Bits)
Short_Float	-3.40282e+38 to 3.40282e+38	32	8	256

Packed_Pulsing_Regime.T:

Thrust pulsing regime selection, packed as an 8-bit enumeration. Values outside the enumeration fail type validation during parameter staging.

Table 15: Packed_Pulsing_Regime Packed Record : 8 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Value	Thr_Firing_Remainder_Enums. Pulsing_Regime.E	0 => On_Pulsing 1 => Off_Pulsing	8	0	7

Field Descriptions:

- **Value** - The thrust pulsing regime.

Parameter.T:

Generic parameter packet for holding a generic parameter

Table 16: Parameter Packed Record : 792 bits (*maximum*)

Name	Type	Range	Size (Bits)	Start Bit	End Bit	Variable Length
Header	Parameter_Header.T	-	24	0	23	-
Buffer	Parameter_Types. Parameter_Buffer_Type	-	768	24	791	Header.Buffer_Length

Field Descriptions:

- **Header** - The parameter header
- **Buffer** - A buffer that contains the parameter type

Parameter_Header.T:

Generic parameter header for holding arbitrary parameters

Table 17: Parameter_Header Packed Record : 24 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Id	Parameter_Types. Parameter_Id	0 to 65535	16	0	15
Buffer_Length	Parameter_Types. Parameter_Buffer_ Length_Type	0 to 96	8	16	23

Field Descriptions:

- **Id** - The parameter identifier
- **Buffer_Length** - The number of bytes used in the parameter type buffer

Parameter_Update.T:

A record intended to be used as a provide/modify connector type for updating/fetching parameters.

Table 18: Parameter_Update Packed Record : 824 bits (*maximum*)

Name	Type	Range	Size (Bits)	Start Bit	End Bit	Variable Length
Table_Id	Parameter_Types. Parameter_ Table_Id	0 to 65535	16	0	15	–
Operation	Parameter_Enums. Parameter_ Operation_ Type.E	0 => Stage 1 => Update 2 => Fetch 3 => Validate	8	16	23	–
Status	Parameter_Enums. Parameter_ Update_ Status.E	0 => Success 1 => Id_Error 2 => Validation_Error 3 => Length_Error	8	24	31	–
Param	Parameter. T	-	792	32	823	–

Field Descriptions:

- **Table_Id** - The ID for the table that contains this parameter
- **Operation** - The parameter operation to perform.
- **Status** - The parameter return status.
- **Param** - The parameter that has been updated or fetched.

Sys_Time.T:

A record which holds a time stamp using EMA VTC format including seconds and subseconds since epoch (1-5-1980 to 1-6-1980 midnight).

Table 19: Sys_Time Packed Record : 48 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Seconds	Interfaces. Unsigned_32	0 to 4294967295	32	0	31
Subseconds	Interfaces. Unsigned_16	0 to 65535	16	32	47

Field Descriptions:

- **Seconds** - The number of seconds elapsed since epoch.
- **Subseconds** - The number of $1/(2^{16})$ sub-seconds.

Thr_Force_Cmd.T:

Table 20: Thr_Force_Cmd Packed Record : 256 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Thr_Force	Packed_F32x8.T	-	256	0	255

Field Descriptions:

- **Thr_Force** - [N] Thruster force values (8 thrusters)

Thr_On_Time_Cmd.T:

Table 21: Thr_On_Time_Cmd Packed Record : 256 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
On_Time_Request	Packed_F32x8.T	-	256	0	255

Field Descriptions:

- **On_Time_Request** - [s] Thruster on-time requests (8 thrusters)

Tick.T:

The tick datatype used for periodic scheduling. Included in this type is the Time associated with a tick and a count.

Table 22: Tick Packed Record : 80 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Time	Sys_Time.T	-	48	0	47
Count	Interfaces. Unsigned_32	0 to 4294967295	32	48	79

Field Descriptions:

- **Time** - The timestamp associated with the tick.
- **Count** - The cycle number of the tick.

5.3 Enumerations

The following section outlines any enumerations used in the component.

Data_Product_Enums.Fetch_Status.E:

This status denotes whether a data product fetch was successful.

Table 23: Fetch_Status Literals:

Name	Value	Description
Success	0	The data product was returned successfully.
Not_Available	1	No data product is yet available for the provided id.
Id_Out_Of_Range	2	The data product id was out of range.

Parameter_Enums.Parameter_Operation_Type.E:

This enumeration lists the different parameter operations that can be performed.

Table 24: Parameter_Operation_Type Literals:

Name	Value	Description
Stage	0	Stage the parameter.
Update	1	All parameters are staged, it is ok to update all parameters now.
Fetch	2	Fetch the parameter.
Validate	3	Validate all the parameters.

Parameter_Enums.Parameter_Update_Status.E:

This status enumeration provides information on the success/failure of a parameter operation.

Table 25: Parameter_Update_Status Literals:

Name	Value	Description
Success	0	Parameter was successfully staged.
Id_Error	1	Parameter id was not valid.
Validation_Error	2	Parameter values were not successfully validated.
Length_Error	3	Parameter length was not correct.

Thr_Firing_Remainder_Enums.Pulsing_Regime.E:

Thrust pulsing regime selection. Values match the C ThrFiringRemainderPulsingRegime enumeration.

Table 26: Pulsing_Regime Literals:

Name	Value	Description
On_Pulsing	0	Thruster on-pulsing regime.
Off_Pulsing	1	Thruster off-pulsing regime.